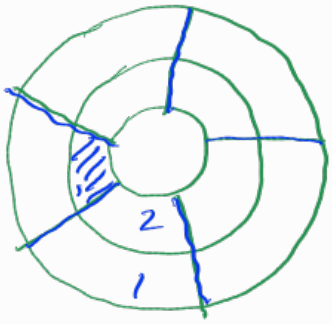
B Tree

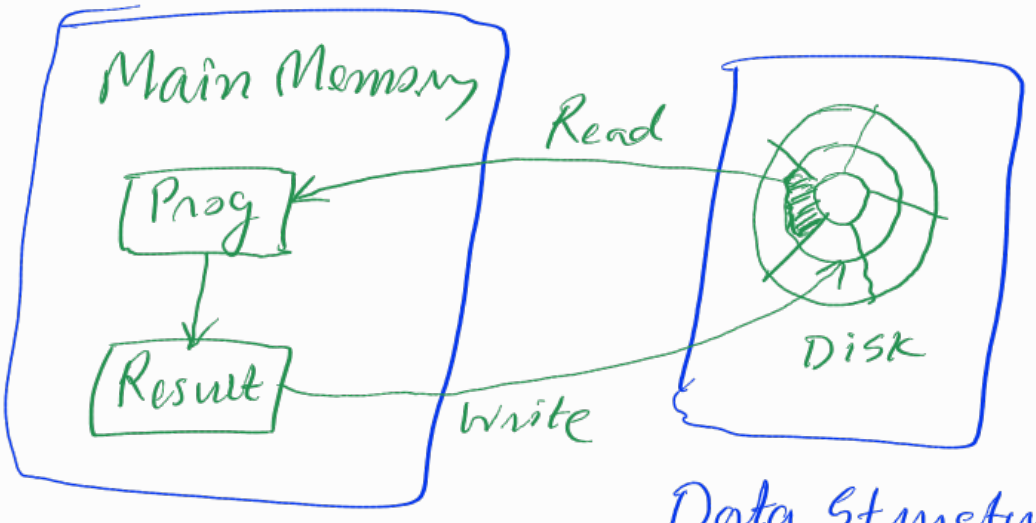
Disk Structure



secton S
track T

$(T+S)$ + offset

Block 512 bytes



RAM
Data Structure (in RAM)

Data Structure in Disk (i.e DBMS)

1. How records are stored? (on Disk)

Students (ID, Name, Dept, Sec. Addrss)

SID = 10 bytes
Name = 50 "
Dept = 10 "
Sec. = 8
Addrss = 50
Total: 128 bytes (each Record)

We've 100 records.

No of records in each Block = $\frac{512}{128}$

$\simeq 4$

So, 100 records, No of Blocks = $\frac{100}{4}$

= 25 blocks

select Dept from students where ID = 201;

All 25 blocks Read

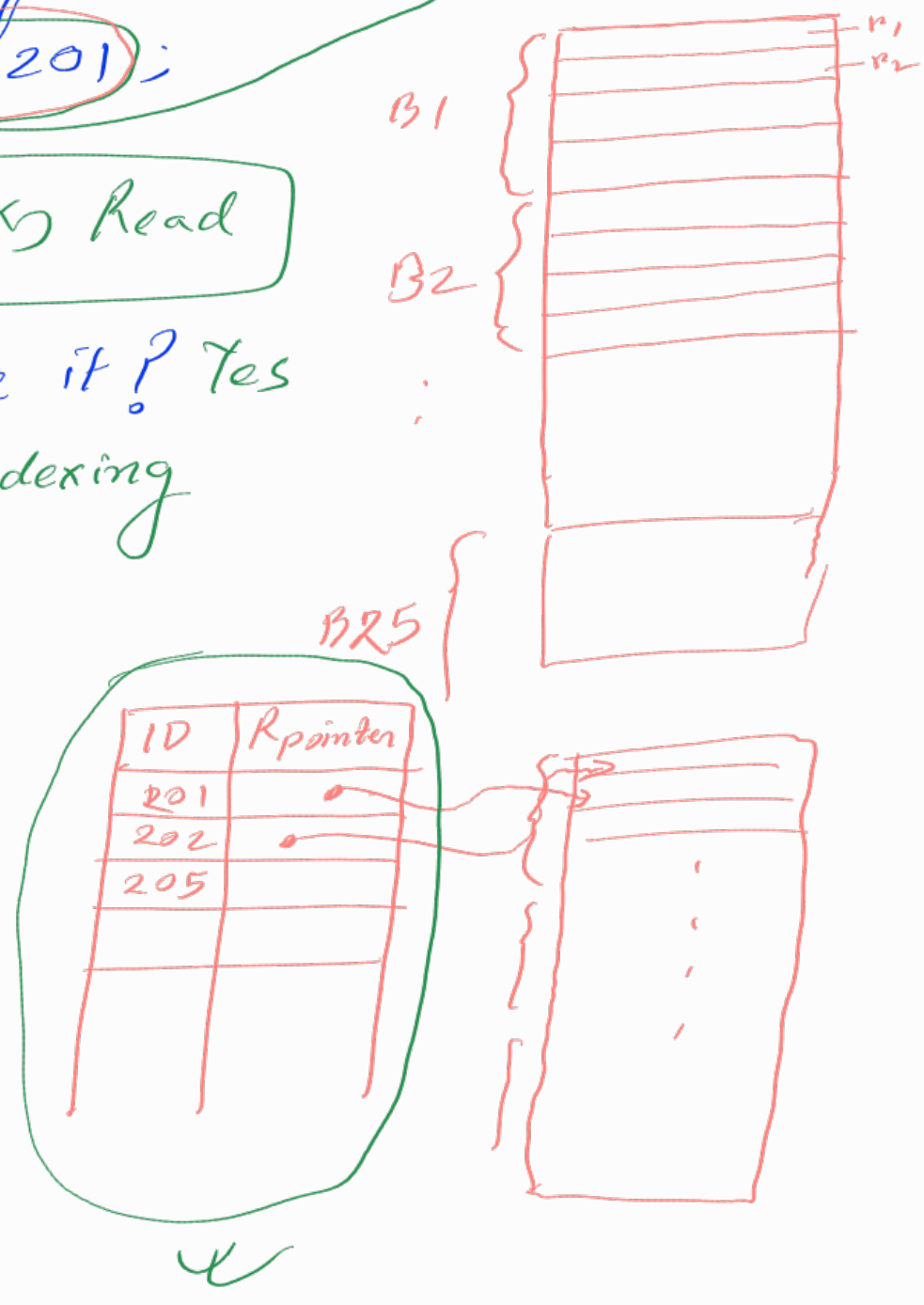
Can we reduce it? Yes

How? By indexing

ID = 10 bytes

Rp = 6 bytes

Total: 16 bytes



No of entries per Block = $\frac{512}{16}$

$= 32$

For 100 records No of Blocks = $\frac{100}{32}$

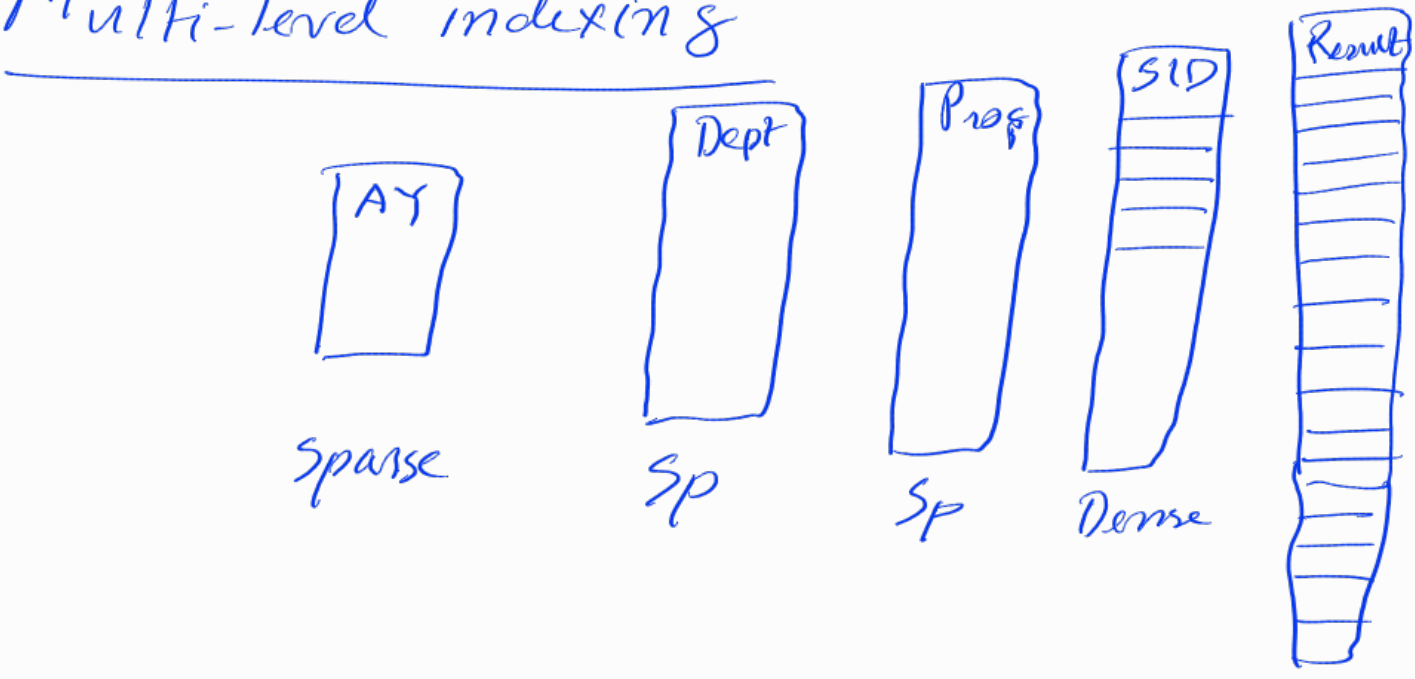
$= 3.2 \simeq 4$ blocks

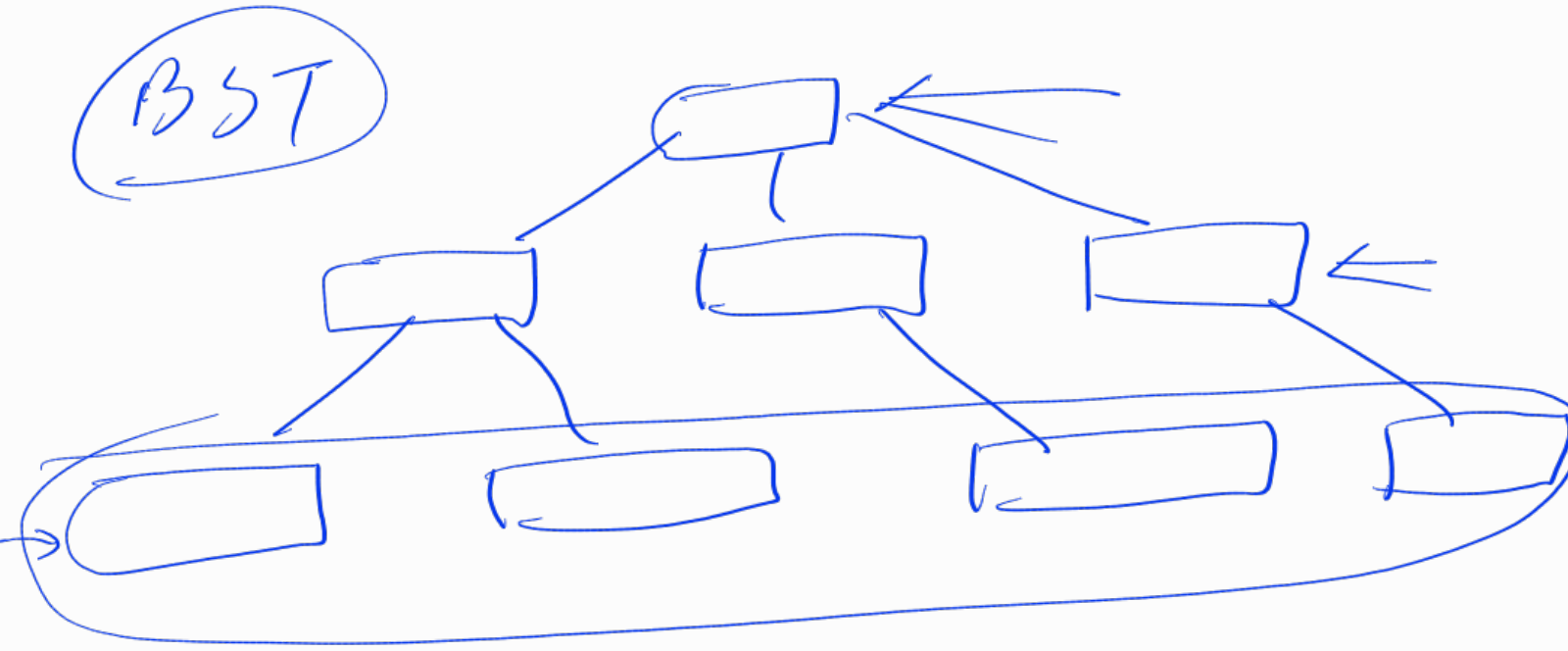
Without indexing $=$ 25 blocks

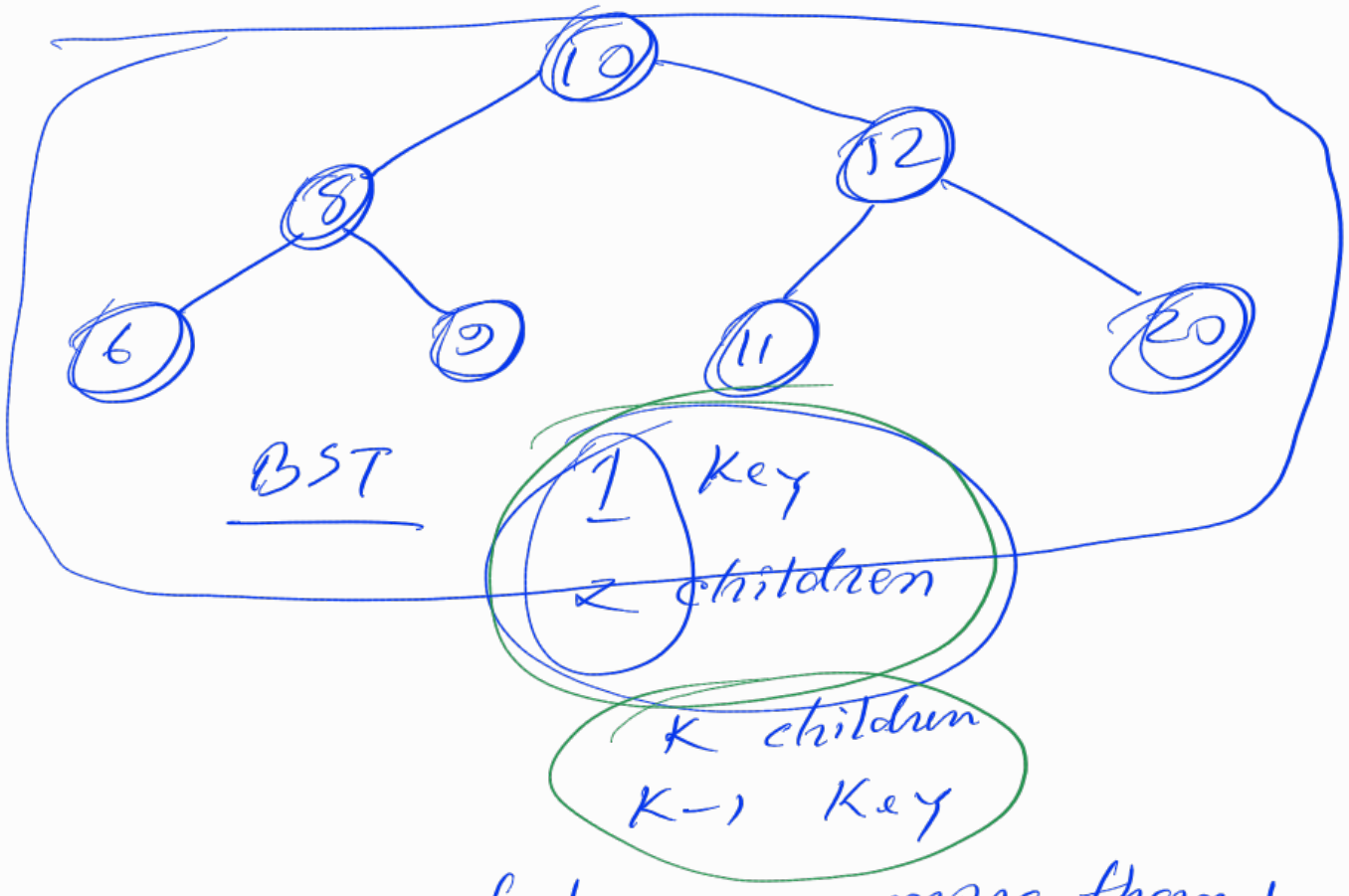
With indexing $=$ (4+1) blocks

5 blocks

Multi-level indexing



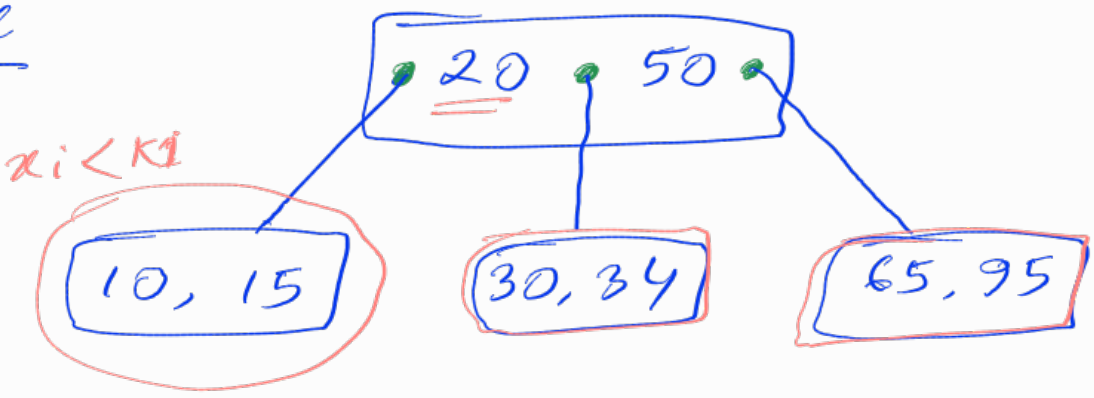




- m-way search tree ⟶ more than 1 key, more than 2 children

$K_1 < K_2 < K_3 \cdots < K_n$

Example

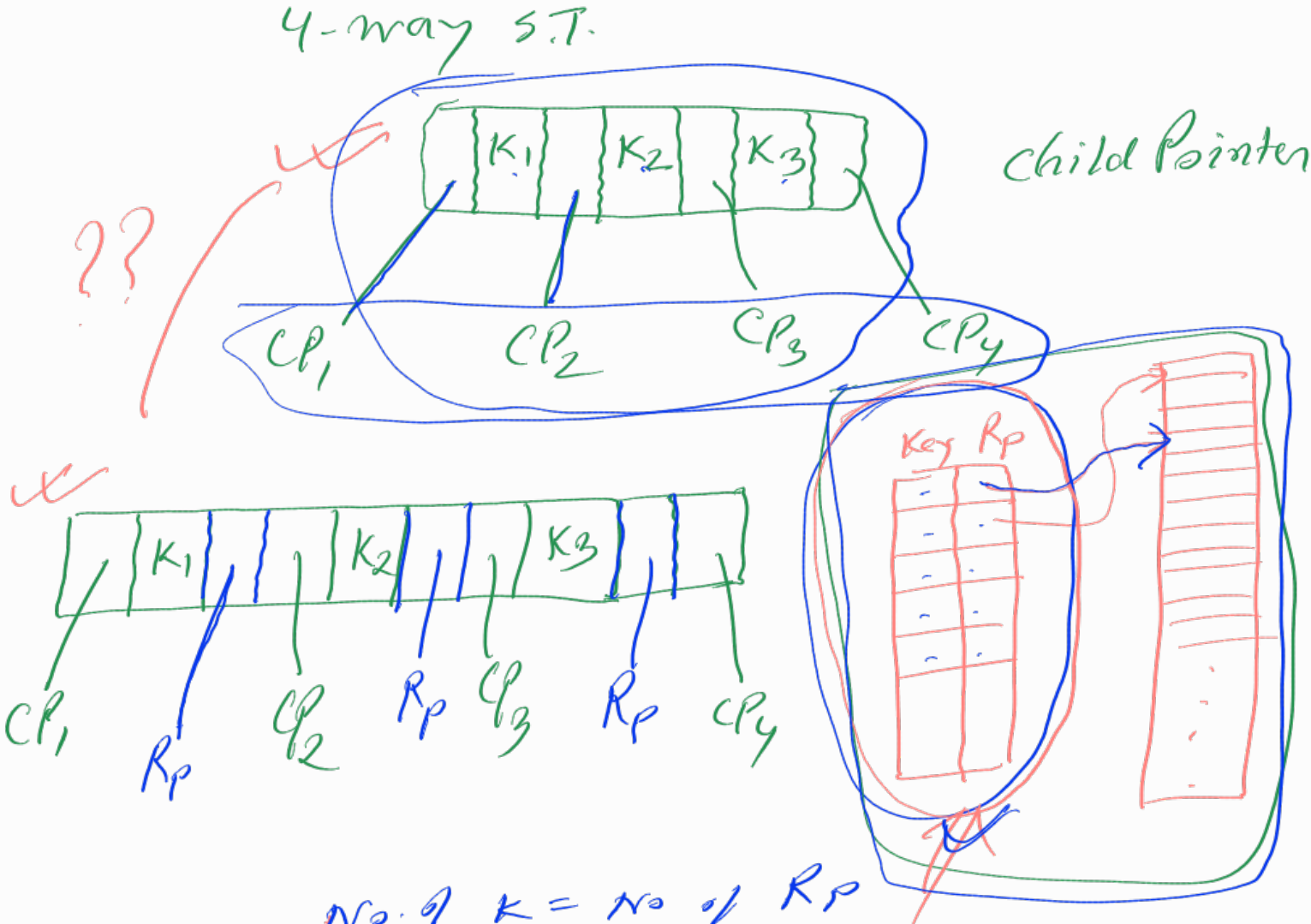


2 Keys
3 children

In general, m-way search tree

(m-1) Keys

this is a 3-way search tree.

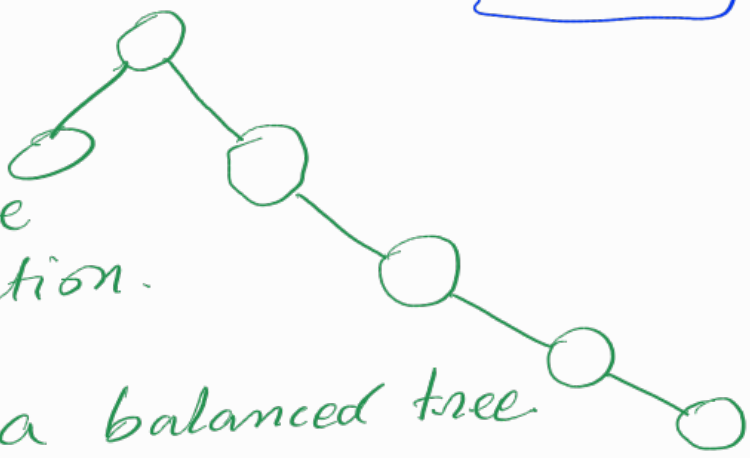


No. of K = No of Rp

Here,	3	Keys
4	cp
3	Rp

m-way S.T.

Problem



It grows in one direction.

It is NOT a balanced tree

Soln: B Tree = m-way ST + Rules

Rules

1. Every node $\lceil \frac{m}{2} \rceil$ children
2. Root can have 2 children (min)
3. All leaves are at the same level (i.e. Balanced)
4. Construction is done by Bottom-up



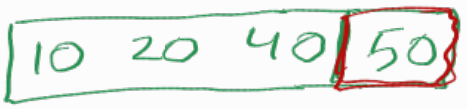
Example

m-way ST

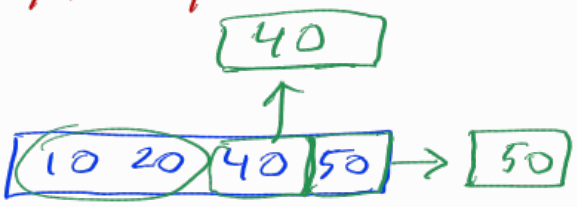
m = 4

K = 4 - 1 = 3 (max size of each node)

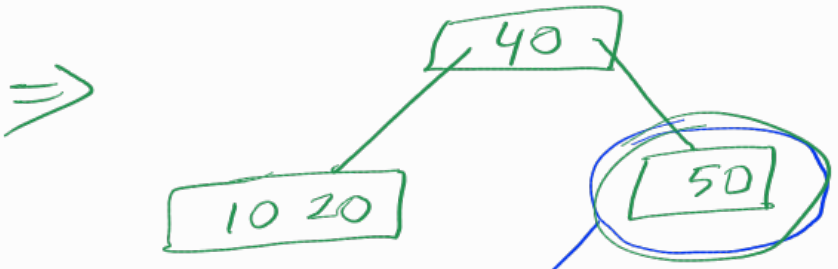
t_1: 10 20 40 50



No space

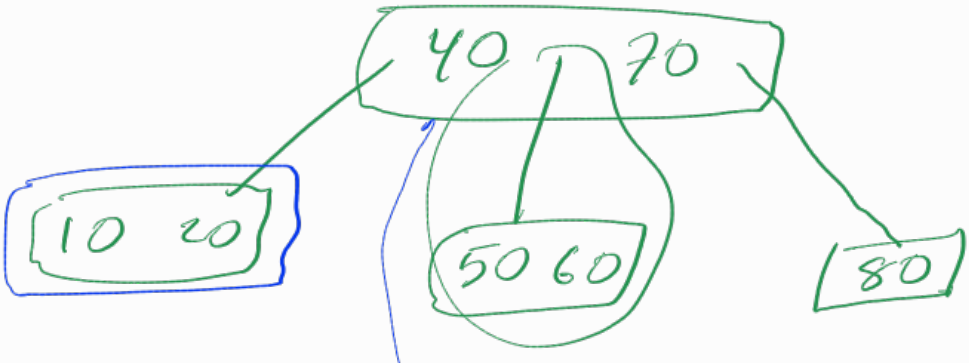


so, go for spliting

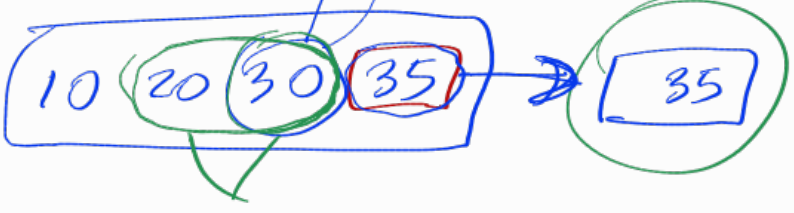


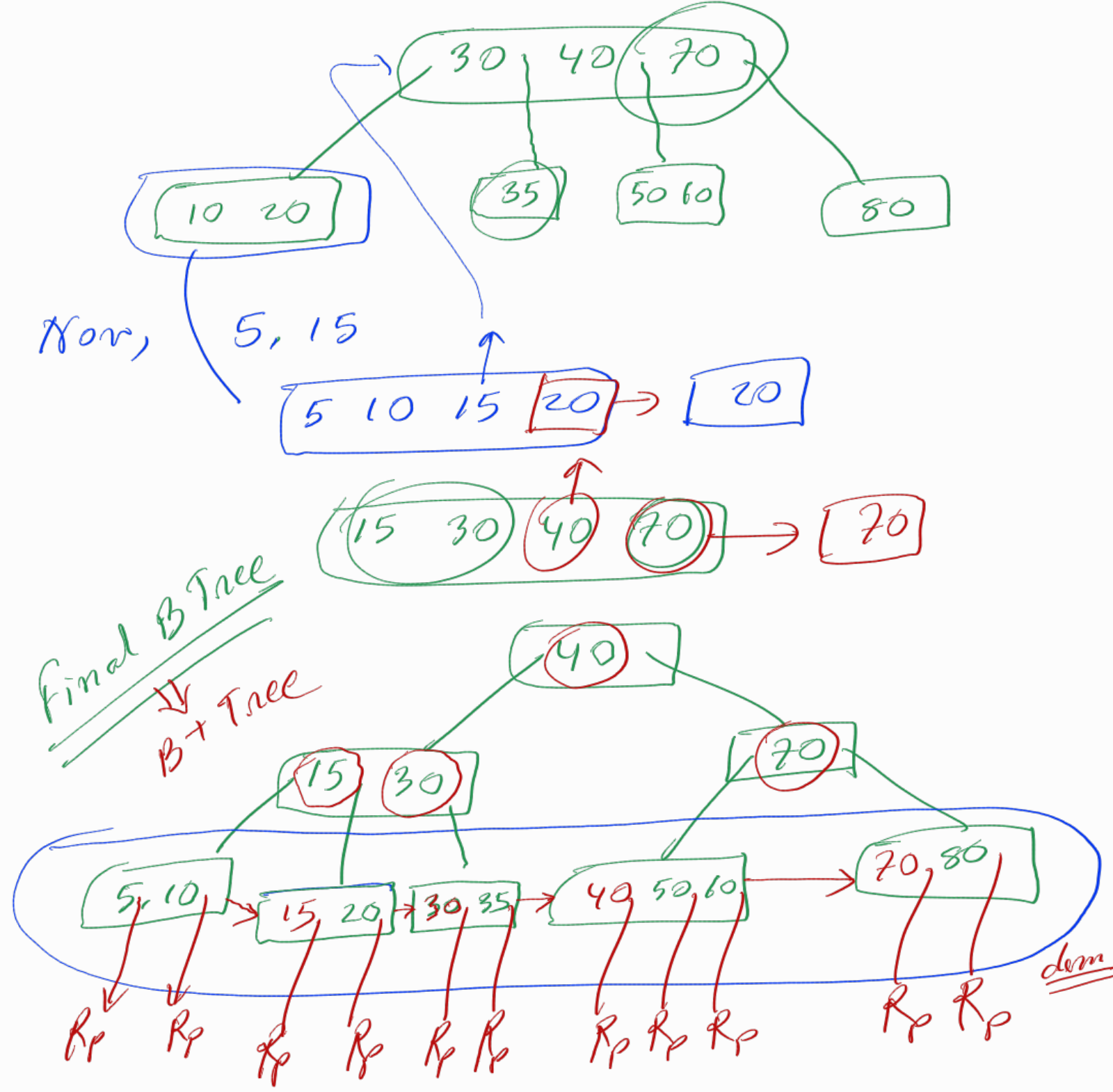
t_2: 60, 70, 80





Now, 30, 35




30 40 70
10 20
35
50 60
80
Now, 5, 15
5 10 15 20
20
15 30 40 70
70
Find B Tree
B+ Tree
40
15 30
20
5, 10
15, 20
30, 35
40, 50, 60
70, 80
Rp
dem